

Lecture 21

6th Nov

ω : n^{th} primitive root of unity

$$\omega^n = 1$$

$$\omega^i \neq 1 \text{ for } 1 \leq i < n$$

Fast Fourier Transform (FFT)

$$\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{(n-1)2} & \dots & \omega^{(n-1)^2} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} P(1) \\ P(\omega) \\ \vdots \\ P(\omega^{n-1}) \end{bmatrix}$$

$$\left[V(\omega) \right]^{-1} = \frac{1}{n} V\left(\frac{1}{\omega}\right)$$

FFT ($n, a_0, a_1, \dots, a_{n-1}, \omega, \text{var } V$)

begin

if $n = 1$ then

$$V[0] = a_0$$

else

FFT ($n/2, a_0, a_2, \dots, a_{n-2}, \omega^2, V$)

FFT ($n/2, a_1, a_3, \dots, a_{n-1}, \omega^2, W$)

for $j = 0$ to $n/2 - 1$ do

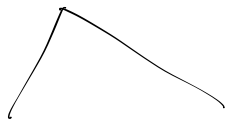
$$V[j] = V[j] + \omega^j W[j]$$

$$V[j + n/2] = V[j] - \omega^j W[j]$$

end

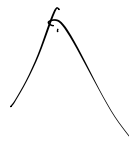
$(0, 1, 2, 3, 4, 5, 6, 7)$

$P_{0,1,2,3,4,5,6,7} (1, \omega, \omega^2, \dots, \omega^7)$



$P_{0,2,4,6} (1, \omega^2, \omega^4, \omega^6)$

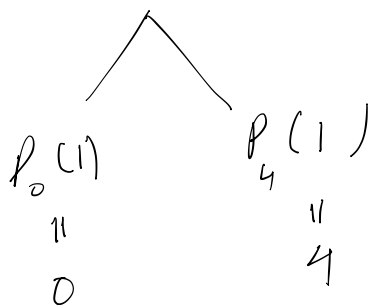
$P_{1,3,5,7} (\omega, \omega^3, \omega^5, \omega^7)$



$P_{0,4} (1, \omega^4)$

$P_{2,6} (1, \omega^4)$

$\vdots$



$\vdots$

eg: $P_{2,6} (\omega^2, \omega^6) = (8, -4)$

$$2 \cdot (\omega^2) + 6 \cdot (\omega^6)$$